

1 The diagram shows a triangle.

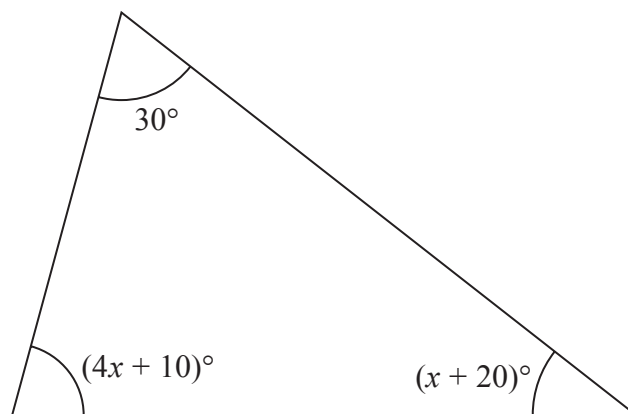


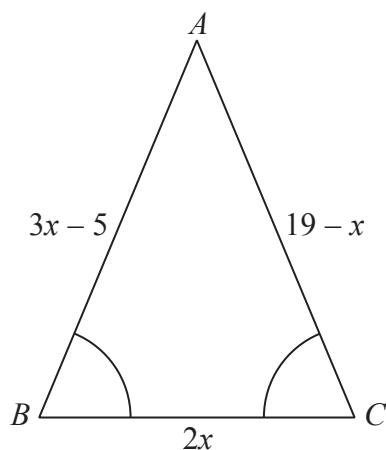
Diagram **NOT**
accurately drawn

Work out the value of x .

$x = \dots\dots\dots$

(Total for Question 1 is 4 marks)

2 ABC is a triangle.



Angle $ABC =$ angle BCA .

The length of side AB is $(3x - 5)$ cm.

The length of side AC is $(19 - x)$ cm.

The length of side BC is $2x$ cm.

Work out the perimeter of the triangle.

Give your answer as a number of centimetres.

..... cm

(Total for Question 2 is 5 marks)

- 3 The diagram shows the triangle PQR .

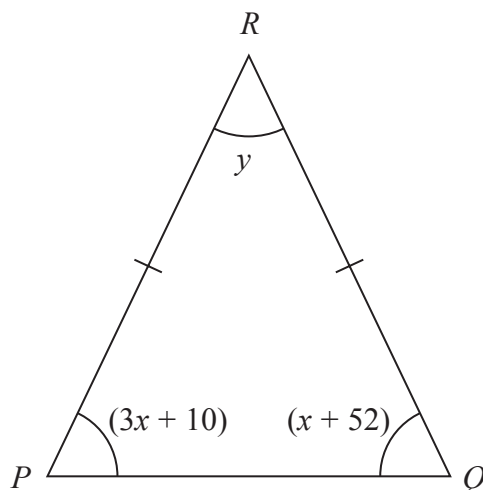


Diagram **NOT**
accurately drawn

In the diagram, all the angles are in degrees.

$$RP = RQ$$

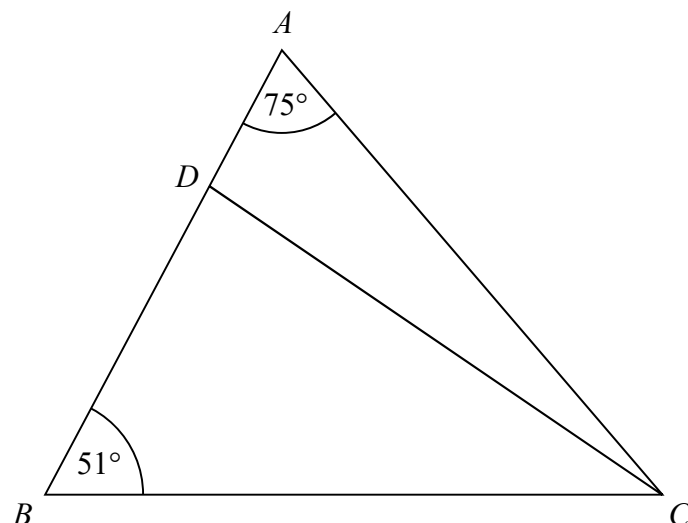
Find the value of y .

Show clear algebraic working.

$$y =$$

(Total for Question 3 is 4 marks)

- 4 The diagram shows triangle ABC .

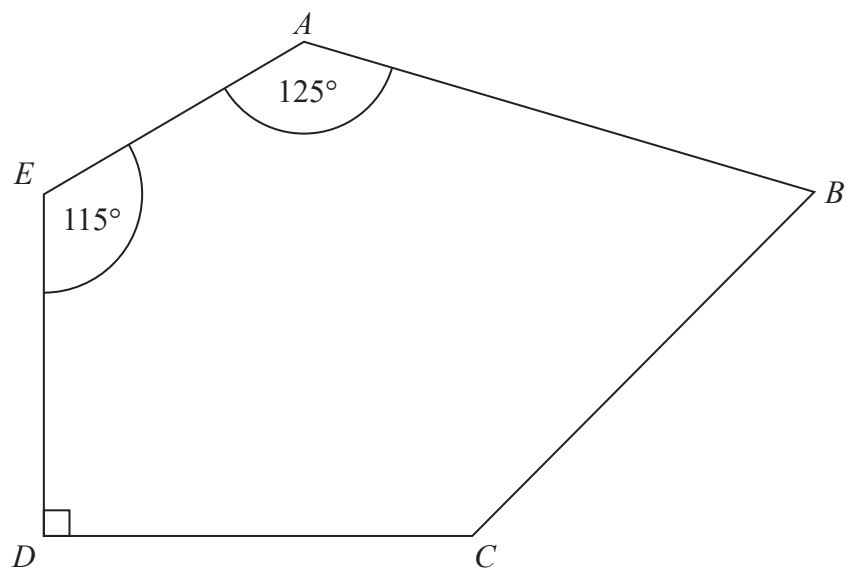


ADB is a straight line.

the size of angle DCB : the size of angle $ACD = 2 : 1$

Work out the size of angle BDC .

5 $ABCDE$ is a pentagon.

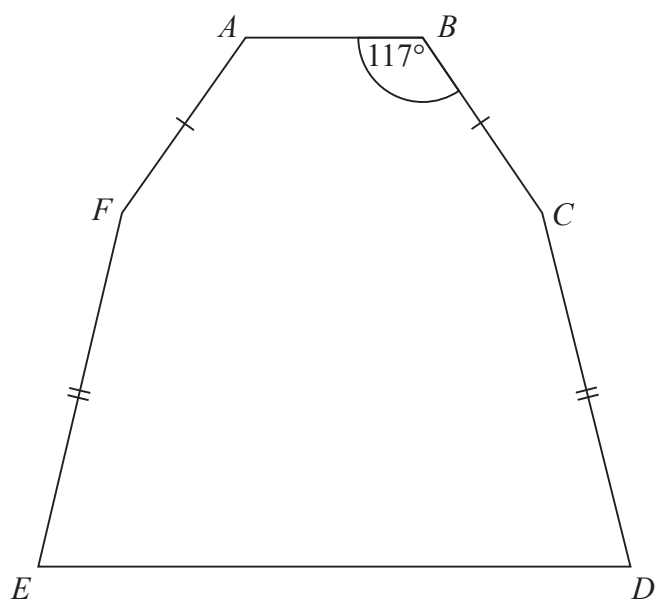


Angle $BCD = 2 \times \text{angle } ABC$

Work out the size of angle BCD .
You must show all your working.

.....
(Total for Question 5 is 5 marks)

- 6 The diagram shows a hexagon.
The hexagon has one line of symmetry.



$$FA = BC$$

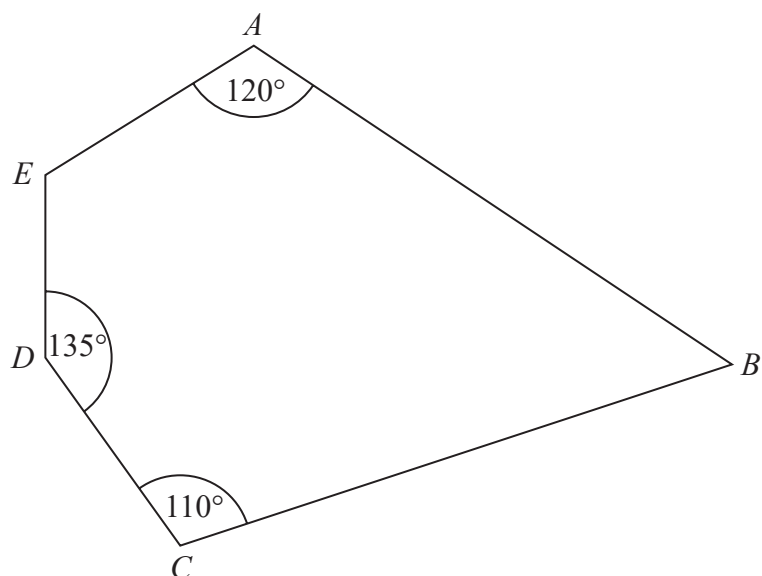
$$EF = CD$$

$$\text{Angle } ABC = 117^\circ$$

$$\text{Angle } BCD = 2 \times \text{angle } CDE$$

Work out the size of angle AFE.
You must show all your working.

7 Here is a pentagon.

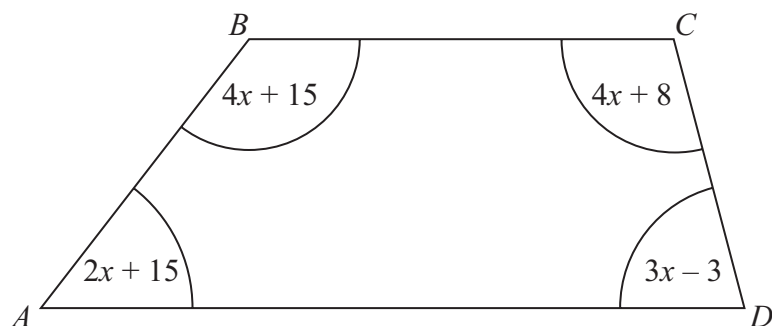


Angle $AED = 4 \times \text{angle } ABC$

Work out the size of angle AED .
You must show all your working.

.....
(Total for Question 7 is 4 marks)

8 $ABCD$ is a quadrilateral.



All angles are measured in degrees.

Show that $ABCD$ is a trapezium.

(Total for Question 8 is 4 marks)

9 $ABCD$ is a trapezium.

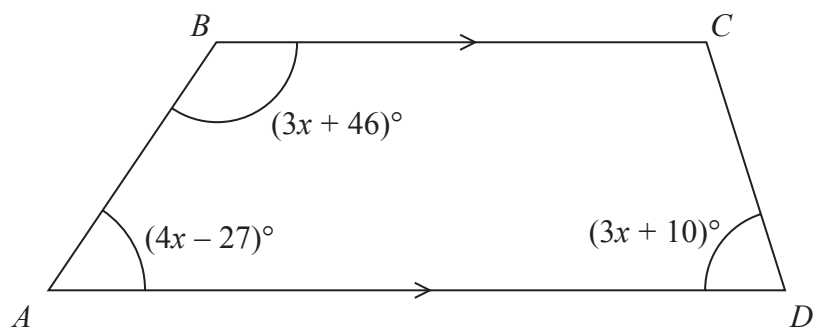


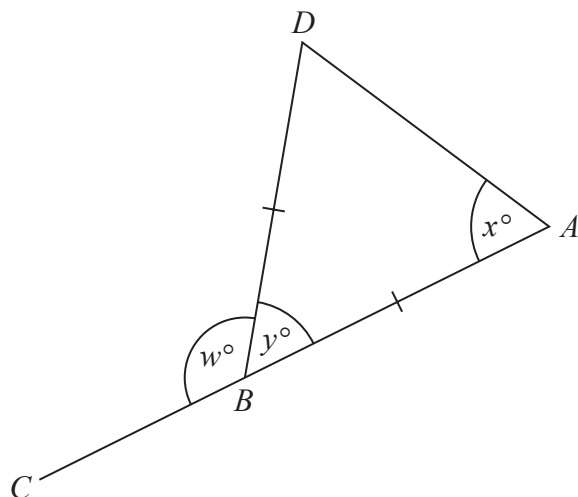
Diagram **NOT**
accurately drawn

BC is parallel to AD

Find the size of the largest angle inside the trapezium.

(Total for Question 9 is 4 marks)

- 10 The diagram shows an isosceles triangle ABD and the straight line ABC .



$$BA = BD$$

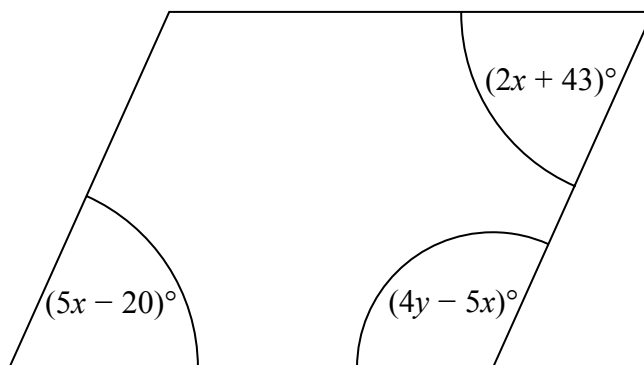
$$x : y = 2 : 1$$

Work out the value of w .

$$w = \dots\dots\dots$$

(Total for Question 10 is 4 marks)

11 Here is a parallelogram.



Work out the value of x and the value of y .

$x =$

$y =$

(Total for Question 11 is 5 marks)